DID EPRC Workshop Talk

Working Slide Outline With Planned Figures and Tables

Draft working document

October 2026

Overview

This document drafts the main content for each slide in the DID EPRC workshop talk. It is not the final slide deck. Each entry gives the slide-facing text and then shows the table, equation, or figure that would appear on that slide.

The empirical motivation is Strickland, Chan, and Gottfried (2026), an *Educational Evaluation and Policy Analysis* paper on NYC's Gifted and Talented (G&T) program using a staggered difference-in-differences design. The workshop exercises use synthetic panel data generated for teaching, inspired by that setting; they are not the NYC administrative data used in the paper.

Part A. DID Setup and Intuition

Slide 1. Title and Roadmap

Difference-in-differences (DID) is a common strategy for studying education policies when some units are exposed to a policy and others are not. The talk starts with the simplest version of DID, then builds toward TWFE, event-study models, and staggered adoption.

Approx. time	Topic
0–5 min	Why DID is useful for education policy questions.
5–15 min	Traditional two-group, two-period DID intuition.
15–25 min	Regression DID, TWFE, and event-study framing.
25–35 min	Staggered treatment timing preview.
35–45 min	Covariates placeholder and handoff to workshop materials.

Planned table for Slide 1: roadmap for the 45-minute talk.

Slide 2. Policy Example: NYC Gifted and Talented

Policy example / research question. NYC's Gifted and Talented (G&T) program. Do students who enter the program experience better outcomes after entry than comparable students who are not yet exposed to G&T? This example is motivated by Strickland, Chan, and Gottfried (2026), which studies the longitudinal effects of NYC's G&T program using a staggered DID design.

Group	Before G&T entry	After G&T entry
Students who enter G&T	Pre-entry outcomes	Post-entry outcomes
Comparison students	Same pre period	Same post period

Planned table for Slide 2: the G&T example as a simple before/after comparison.

Workshop data. The empirical study motivates the policy example, but the workshop uses synthetic teaching data. The generated data keep the same DID logic while simplifying the setting to district-year panels.

Slide 3. What Data Do We Need for DID?

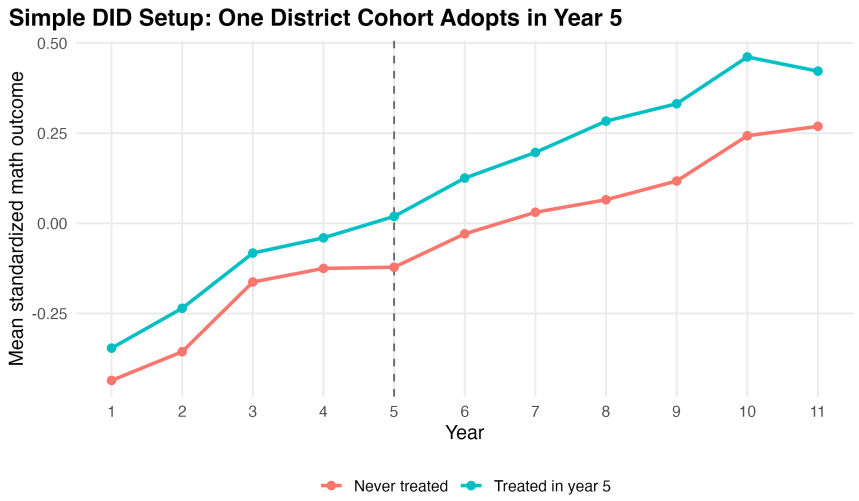
To estimate DID, we need repeated observations on units over time, an outcome, treatment status, and a way to define pre- and post-treatment periods. In the synthetic example, the unit is a district and the policy turns on for treated districts in a known year.

DID ingredient	G&T interpretation	Synthetic data variable
Unit	Student, school, or district	`district_id`
Time	Year or grade	`year`
Outcome	Achievement or attendance	`outcome`
Treatment group	Exposed to G&T or policy	`treated_ever`
Post period	After entry or adoption	`post`
Treatment timing	First exposure year	`treatment_start_year`

Planned table for Slide 3: variables needed for the synthetic DID example.

Slide 4. Traditional DID and Parallel Trends

Key identifying idea. DID asks what would have happened to treated units after treatment if they had not been treated. The comparison group is useful if, before treatment, it was moving in parallel with the treated group. In this simple synthetic example, one group adopts treatment in year 5 and one group is never treated.

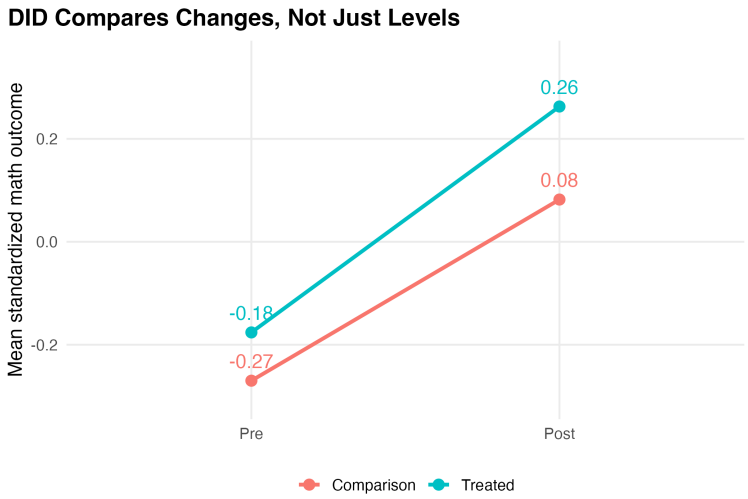


Planned figure for Slide 4: treated units adopt in year 5; comparison units are never treated.

Slide 5. DID Compares Changes, Not Just Levels

The DID estimand subtracts the comparison group's before-after change from the treated group's before-after change. This is why the method is called "difference-in-differences."

$$\widehat{\delta}_{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$



Planned figure for Slide 5: the two-by-two DID comparison.

Slide 6. Regression Version of the Same Idea

The same two-by-two contrast can be estimated with a regression. The coefficient on the interaction between treatment group and post period is the DID estimate. Because the data are generated, we can compare the estimated effect with the known simulated effect.

$$Y_{it} = \alpha + \beta_1 Treat_i + \beta_2 Post_t + \delta(Treat_i \times Post_t) + \varepsilon_{it}.$$

Model	Estimated effect	Known simulated effect
Two-by-two DID regression	0.087	0.090

Planned table for Slide 6: estimated DID effect versus the known simulated effect.

Part B. TWFE and Event Studies

Slide 7. Two-Way Fixed Effects

Many applied DID papers estimate a two-way fixed effects (TWFE) model. Unit fixed effects absorb stable differences across units, and year fixed effects absorb shocks common to all units in a given year. In the simple one-treatment-cohort case, TWFE maps closely onto the standard DID contrast.

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}.$$

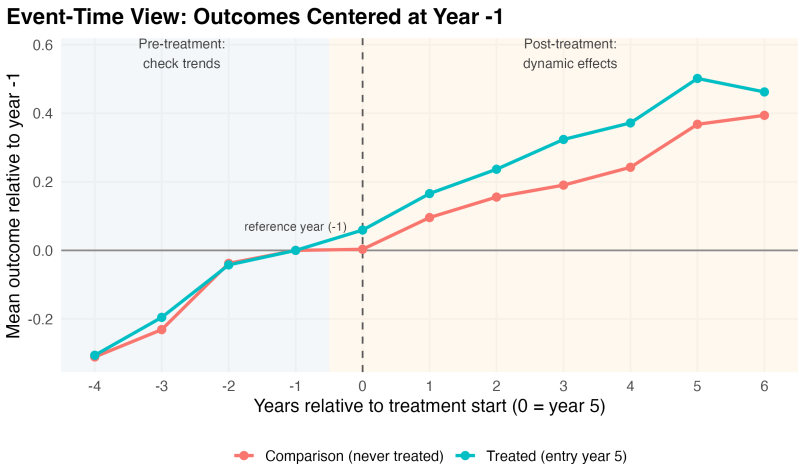
Term	Interpretation
α_i	Unit fixed effects: stable differences across units.
λ_t	Year fixed effects: shocks common to all units in a year.
D_{it}	Indicator that turns on after treatment starts.
δ	Average treatment effect estimate in the simple setup.

Planned table for Slide 7: pieces of the TWFE model.

Slide 8. Event Studies: Re-centering Time Around Treatment

Event studies extend DID by comparing outcomes at multiple time points before and after treatment. The key move is to re-center time around treatment entry. In the G&T example, event time 0 is the year a student enters G&T; event time -1 is the year just before entry. In the simple synthetic data, year 5 is treatment start, so year 4 is the reference year.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k 1[t - G_i = k] + \varepsilon_{it}.$$



Planned figure for Slide 8: outcomes centered at event time -1. Pre-treatment event times show whether treated and comparison groups were already diverging; post-treatment event times show how effects evolve after treatment.

Slide 9. What a Single TWFE Coefficient Can Hide

A single TWFE treatment coefficient can be too compressed if effects change over time or if the design includes multiple treatment timings. The event-study setup helps make those dynamics visible before moving into the staggered DID case.

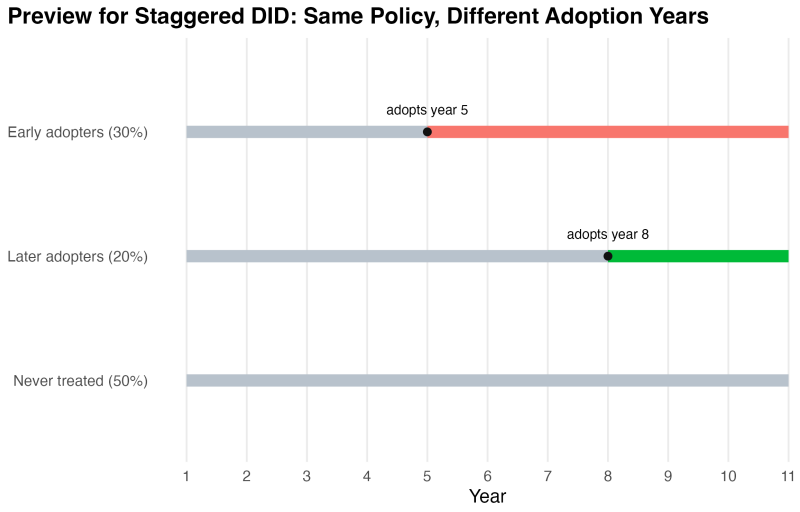
What we estimate	What it can miss
One TWFE treatment coefficient	Whether effects grow, fade, or appear only after several years.
Average post-treatment effect	Differences across event time.
Simple treated/control comparison	Whether pre-treatment trends were already diverging.

Planned table for Slide 9: why event-time thinking is useful before staggered DID.

Part C. Staggered DID Preview

Slide 10. Staggered Treatment Timing

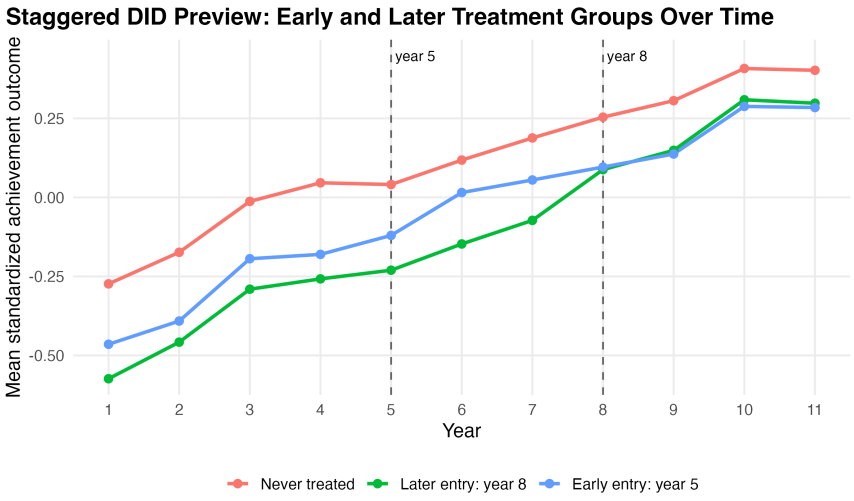
In many education settings, treatment does not start for everyone at the same time. In the G&T example, students can enter the program at different grade points. In the synthetic workshop data, one treated group adopts in year 5 and another treated group adopts in year 8.



Planned figure for Slide 10: early adopters, later adopters, and never-treated units.

Slide 11. Group Trends With Two Treated Cohorts

With staggered timing, the comparison problem changes. In the same calendar year, some units may be newly treated, some may already be treated, and some may not yet be treated. This slide previews the issue without fully covering the staggered DID estimators.



Planned figure for Slide 11: early-treated, later-treated, and never-treated group trends.

Slide 12. Covariates Placeholder

Placeholder for Wei's section. This slide will introduce the question of covariates in DID: what changes when parallel trends is framed as conditional parallel trends, and how should time-varying versus time-invariant covariates be handled in TWFE and staggered DID estimators?

Issue	Placeholder content
Conditional parallel trends	Define the assumption and connect it to covariate adjustment.
Time-invariant covariates	Clarify what unit fixed effects already absorb.
Time-varying covariates	Flag why timing and interpretation require care.
Estimator choice	Add CSA/Wooldridge/Sant'Anna and Zhao details later.

Planned table for Slide 12: placeholder structure for the covariates discussion.

Slide 13. Handoff and Workshop Materials

The introductory section closes by pointing participants to the workshop materials: the synthetic data, starter code, outline/slides, and readings.

Material	Where it lives
Outline/slides	Workshop website Slides tab.
Synthetic data	Workshop website Data tab; CSV and Stata files.
Code	Workshop website Code tab; starter R scripts.
Readings	Workshop website Resources tab.

Planned table for Slide 13: final slide with workshop links.

https://katjanestrickland.github.io/did-eprc-workshop/

Reference To Include

Strickland, K., Chan, W., and Gottfried, M. A. (2026). Longitudinal effects of NYC's Gifted and Talented program: Evidence from a staggered difference-in-differences approach. *Educational Evaluation and Policy Analysis*. https://doi.org/10.3102/01623737261438146